

CH4207

Chemistry of Materials

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Syllabus:

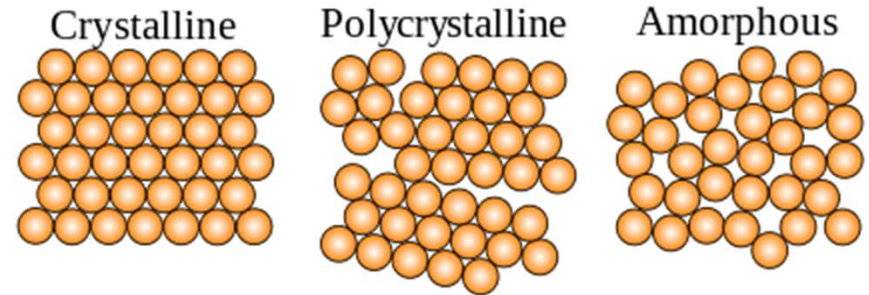
Crystal Structures
Reciprocal Lattice
Symmetry
Electronic Properties of Solids

Marks distribution:

Class test (SB):	15
Class test (SR):	15
Mid-sem:	20
End-sem:	50

Solids

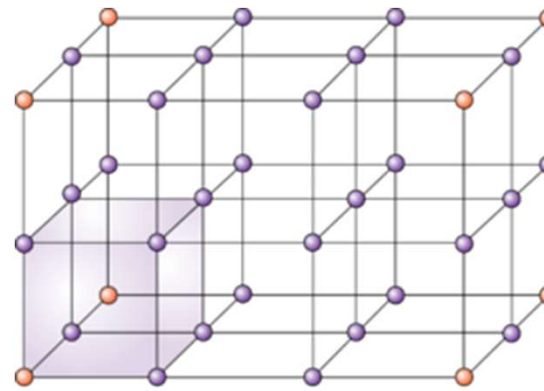
- Solids are composed on atoms & molecules
- Arrangement is in 3-dimensions (3D)
- Arrangement is **periodic** in nature



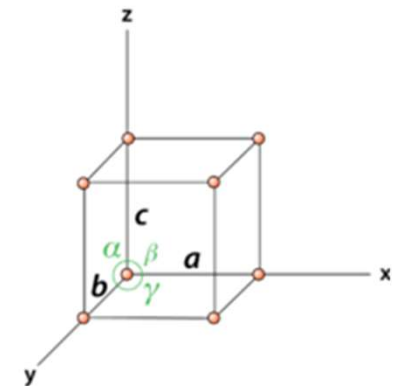
Crystals of diff. size
& orientation random

Arrangement of atoms in a crystal

- Building block of each crystal : **Unit Cell**
-Unit cell is the smallest cube that represents the greater geometry of a larger crystal
- The structure of all crystals can be described in terms of a **lattice**; with a group of atoms attached to each lattice point
- The group of atoms is called the **basis**; when repeated in space it forms the **crystal structure**



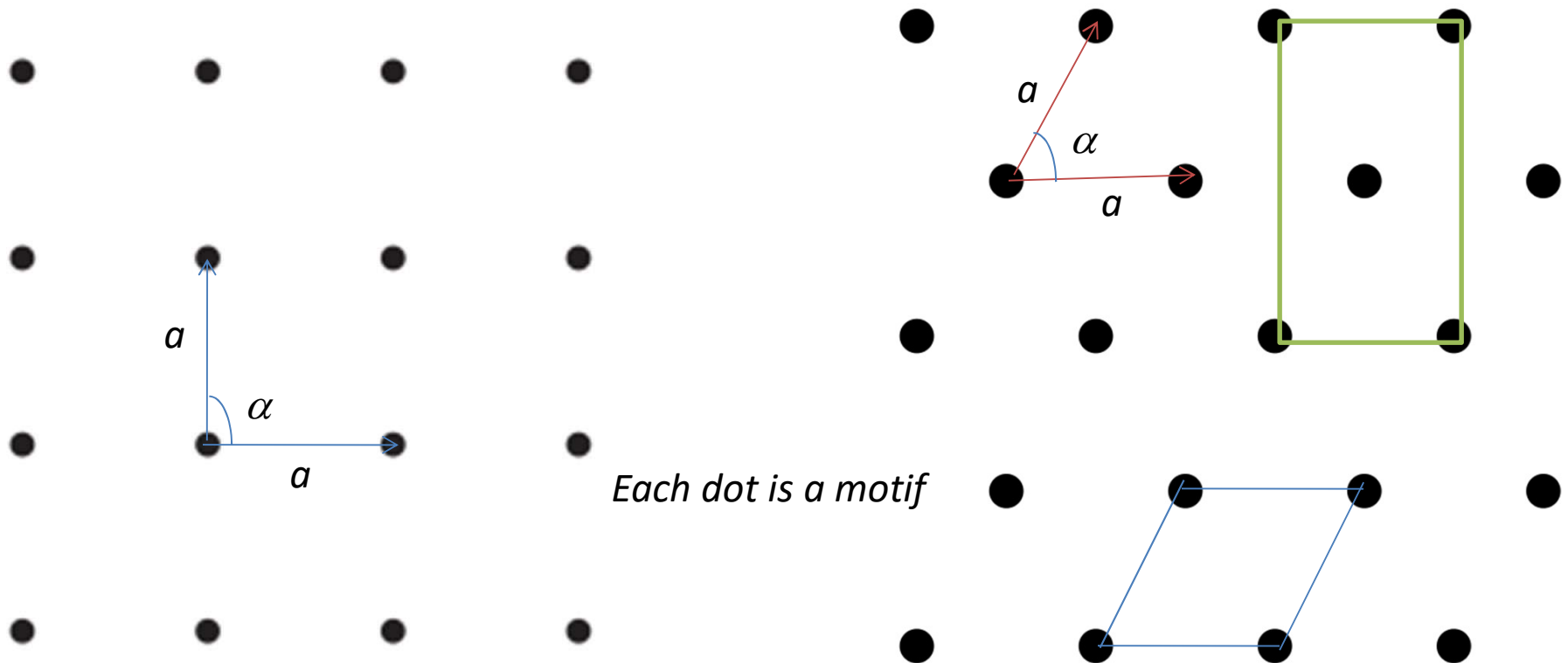
Crystal Lattice



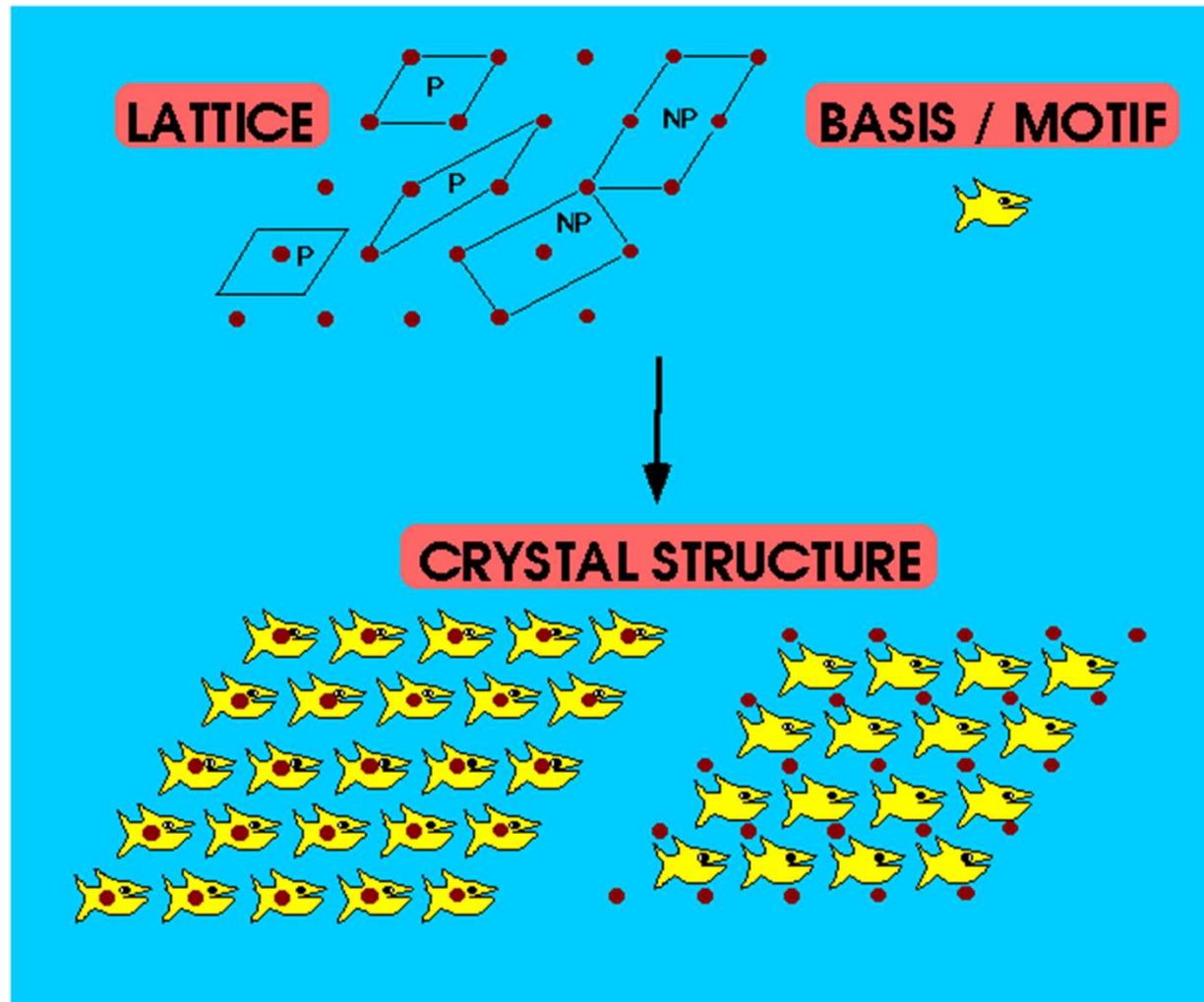
Unit Cell

Solids

- **Basis (motif)** → **What** is the unit that is being repeated?
- Scheme of repetition (**How**) → **Lattice**
- **Lattice + Basis** → **Crystal Structure**



Lattice + Basis (Motif) = Crystal Structure



Lattice + Basis (Motif) = Crystal Structure

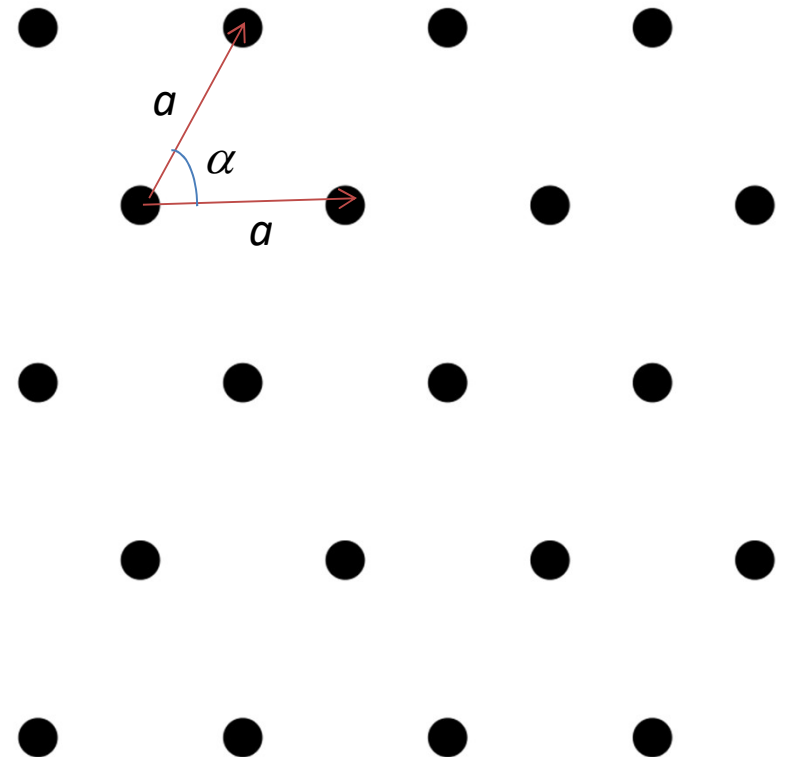
Lattice: objects of identical dimensions

- (1) Arrangement of atoms in regular arrays in 3D
- (2) Environment of each lattice point is equal everywhere
- (3) Interfacial angles are constant

Periodic arrangement

Total arrangement of points in space
→ 14 types (**14 Bravais lattices**)

14 Bravais lattices can be described in
7 types of unit cell
→ **Crystal system**

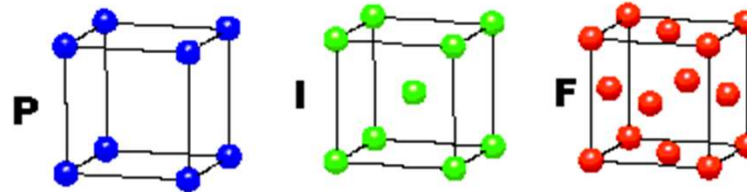


14 Possible Bravais Lattices

CUBIC

$$a = b = c$$

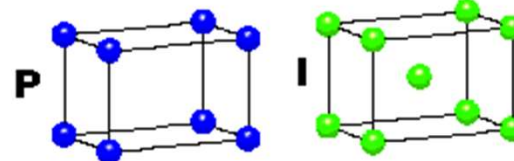
$$\alpha = \beta = \gamma = 90^\circ$$



TETRAGONAL

$$a = b \neq c$$

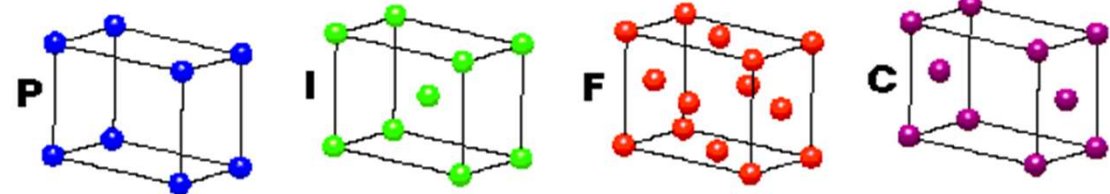
$$\alpha = \beta = \gamma = 90^\circ$$



ORTHORHOMBIC

$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

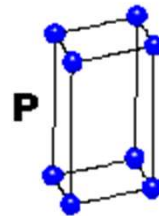


HEXAGONAL

$$a = b \neq c$$

$$\alpha = \beta = 90^\circ$$

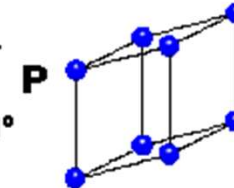
$$\gamma = 120^\circ$$



TRIGONAL

$$a = b = c$$

$$\alpha = \beta = \gamma \neq 90^\circ$$

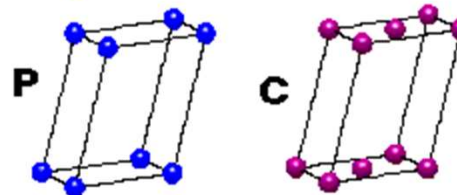


MONOCLINIC

$$a \neq b \neq c$$

$$\alpha = \gamma = 90^\circ$$

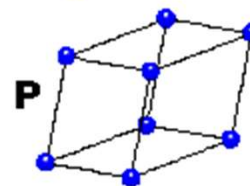
$$\beta \neq 120^\circ$$



TRICLINIC

$$a \neq b \neq c$$

$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$



4 Types of Unit Cell

P = Primitive

I = Body-Centred

F = Face-Centred

C = Side-Centred

+

7 Crystal Classes

→ 14 Bravais Lattices

Counting Atoms in 3D Cells

Atoms in different positions in a cell are shared by differing numbers of unit cells

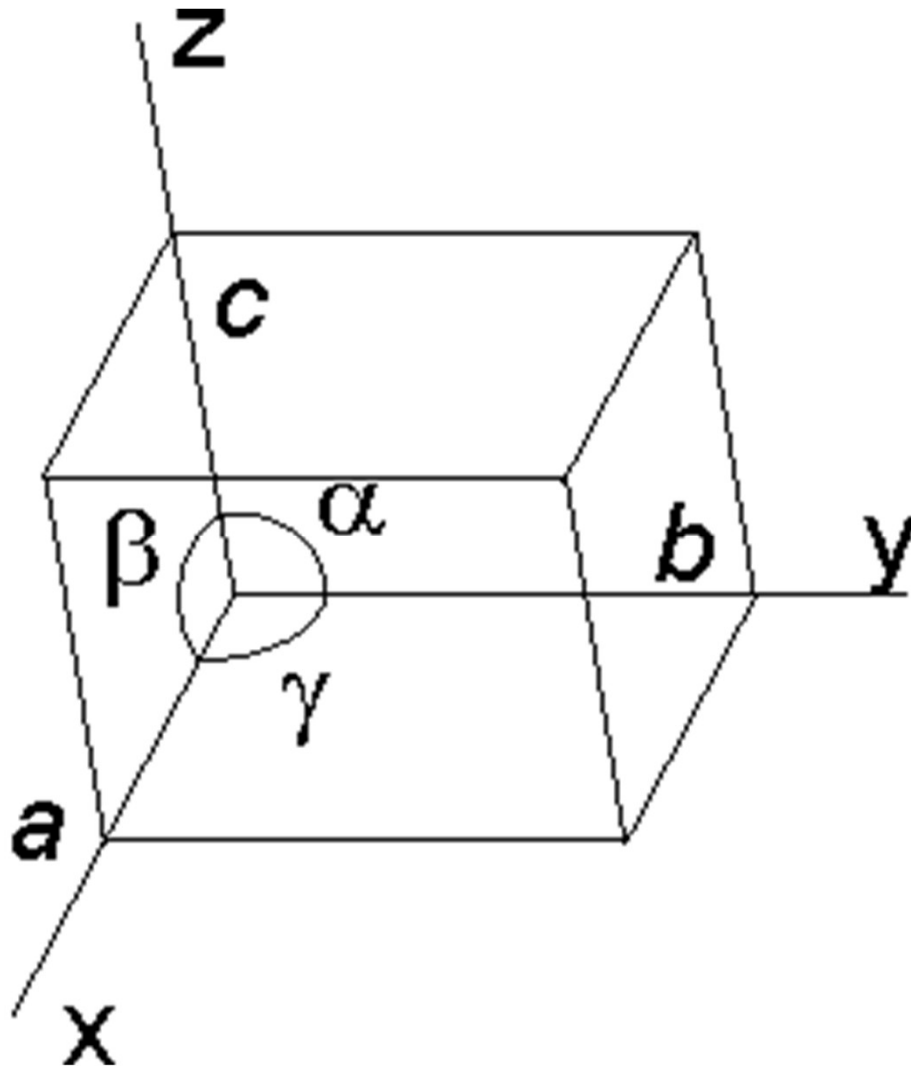
- **Vertex** atom shared by **8** cells $\Rightarrow \frac{1}{8}$ atom per cell
- **Edge** atom shared by **4** cells $\Rightarrow \frac{1}{4}$ atom per cell
- **Face** atom shared by **2** cells $\Rightarrow \frac{1}{2}$ atom per cell
- **Body** unique to **1** cell $\Rightarrow 1$ atom per cell

On combining 7 **Crystal Classes** with 4 possible unit cell types Symmetry indicates that only 14 3-D lattice types occur

Combining these 14 Bravais lattices with all possible symmetry elements

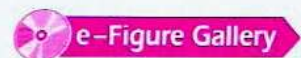
$\longrightarrow$ **230** different **Space Groups**

7 Crystal Systems



Angle made by x and $y \rightarrow \gamma$
Angle made by y and $z \rightarrow \alpha$
Angle made by x and $z \rightarrow \beta$

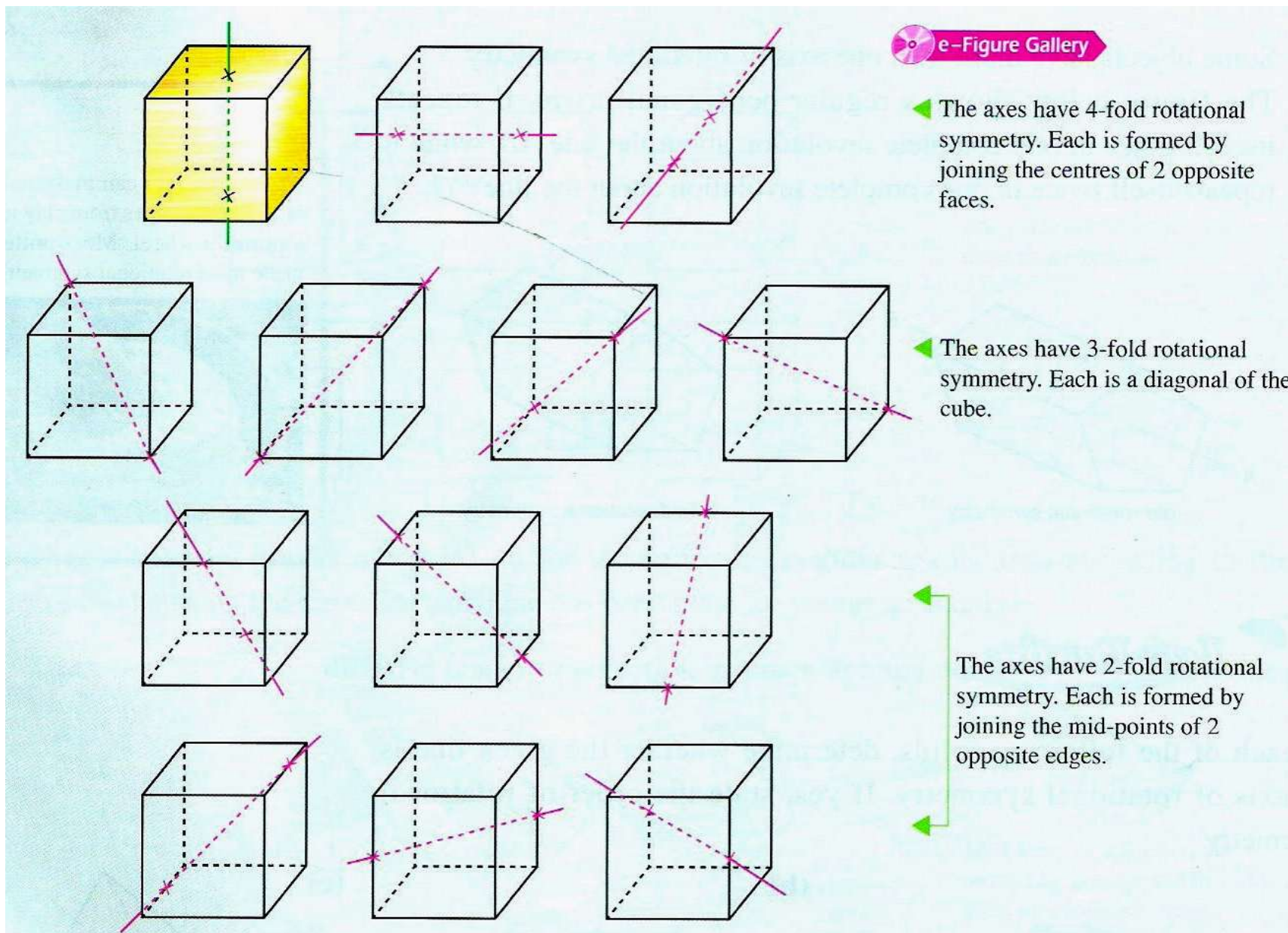
Cube



▶ The axes have 4-fold rotational symmetry. Each is formed by joining the centres of 2 opposite faces.

▶ The axes have 3-fold rotational symmetry. Each is a diagonal of the cube.

▶ The axes have 2-fold rotational symmetry. Each is formed by joining the mid-points of 2 opposite edges.



Understanding the crystal system

Lattice planes

SYMBOL: ()

(hkl) : Miller indices

Entirely imaginary and simply provide a reference grid to which the atoms in the crystal structure may be referred

The planes differ based on the density of atoms on the plane

Intercepts: x/a y/b z/c

Reciprocals: a/x b/x c/z

Miller indices: h k l

(hkl) (001)

Intercepts: ∞/a ∞/b $1/c$

Reciprocals: a/∞ b/∞ $c/1$

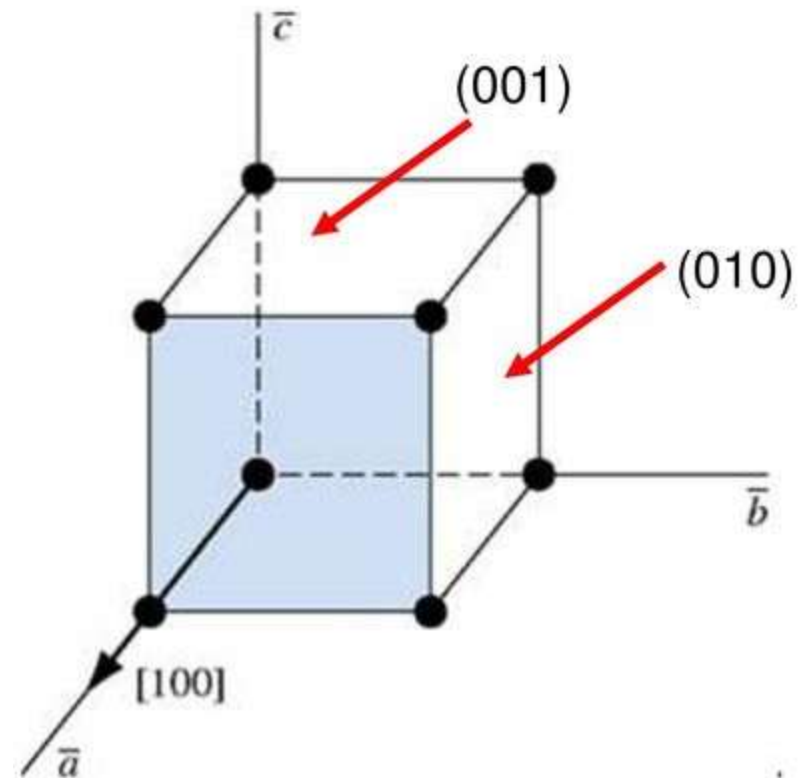
Miller indices: 0 0 1

(hkl) (001)

Intercepts: ∞/a $1/b$ ∞/c

Reciprocals: a/∞ $b/1$ c/∞

Miller indices: 0 1 0

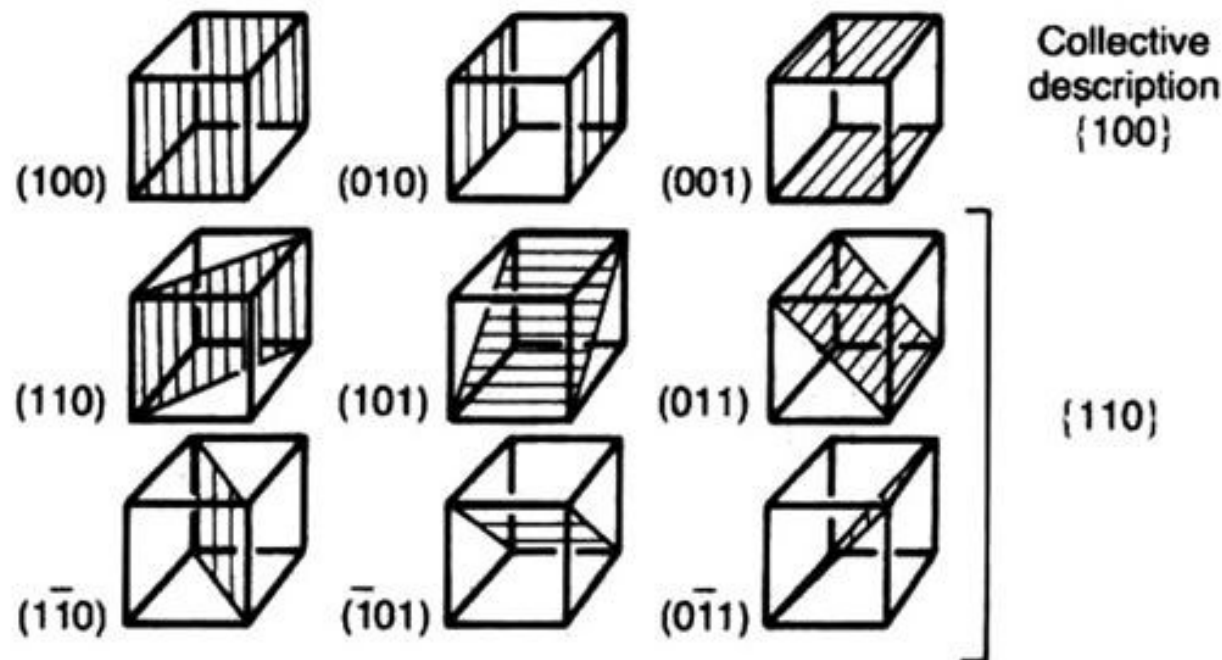


Understanding the crystal system

Equivalent planes { }

Planes will be equivalent when distribution of points on the planes are the same i.e. density of planes are the same

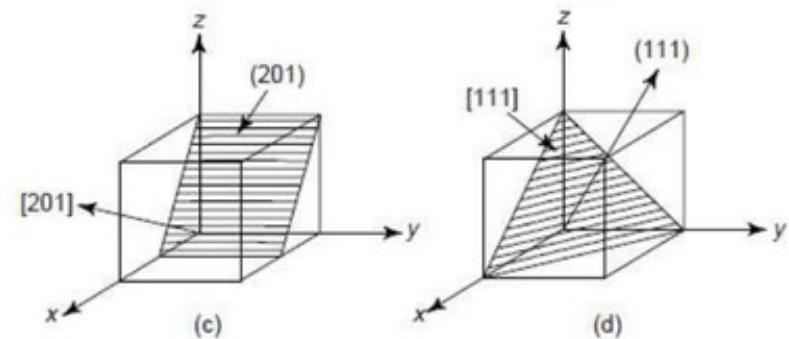
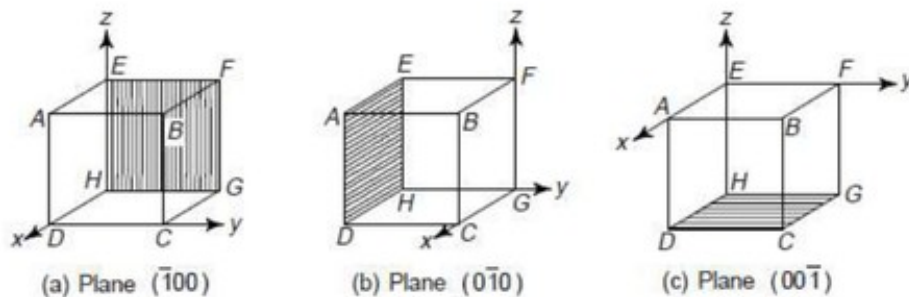
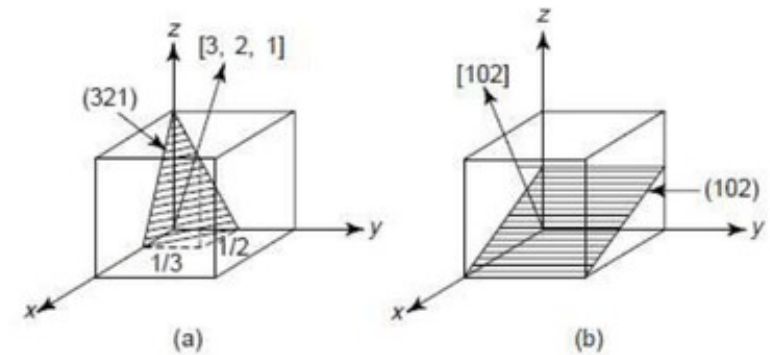
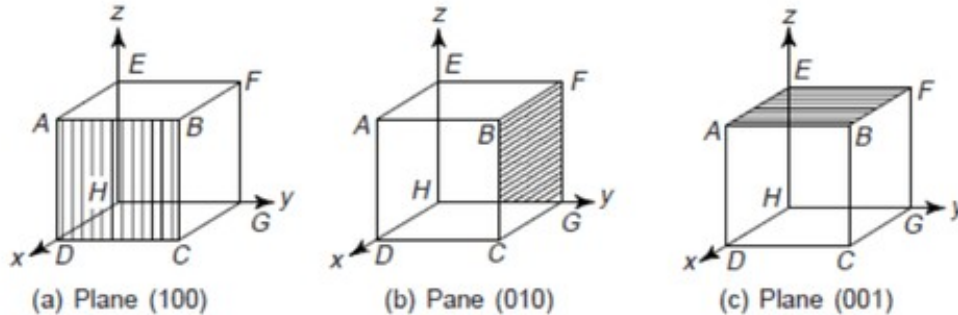
For a simple cube: $\{100\}$ $(100) (010) (001) (\bar{1}00) (0\bar{1}0) (00\bar{1})$



Understanding the crystal system

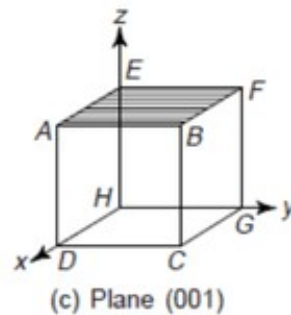
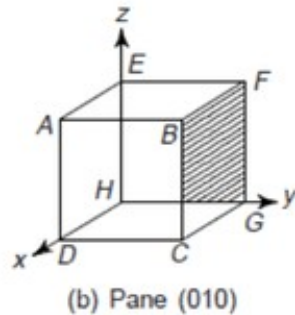
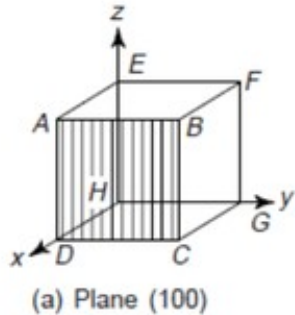
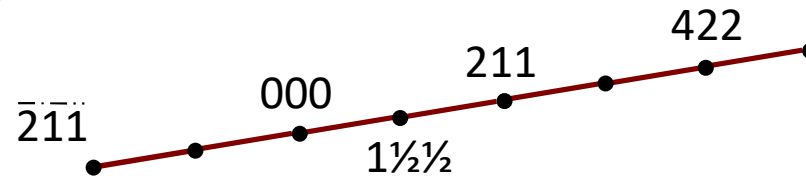
Directions []

[hkl] direction is always perpendicular to (hkl) plane



Understanding the crystal system

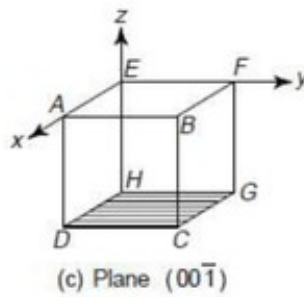
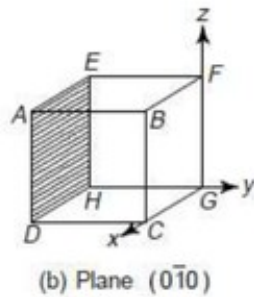
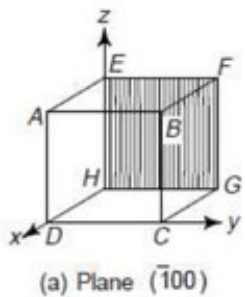
Equivalent directions $\langle \quad \rangle$



$[100]$ $[\bar{1}00]$ Equivalent

$[010]$ $[0\bar{1}0]$ Equivalent

$[001]$ $[00\bar{1}]$ Equivalent



Miller indices in hexagonal system

$(h\ k\ i\ l)$

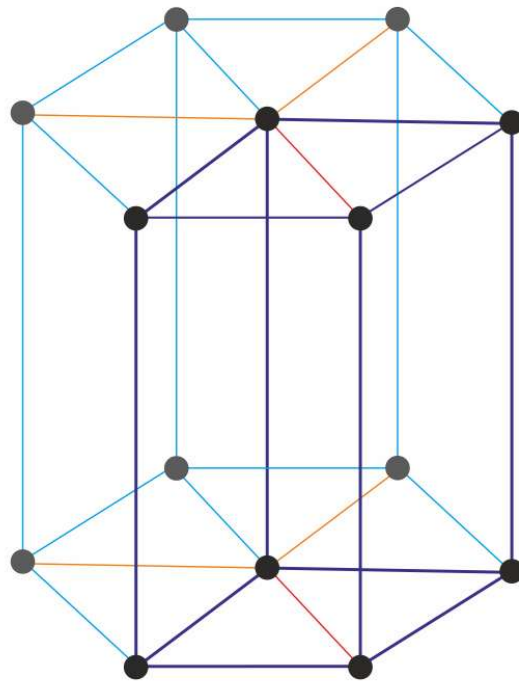
$$h + k + i = 0$$

$$h + k = -i$$

Thus, i -index is a piece of redundant information

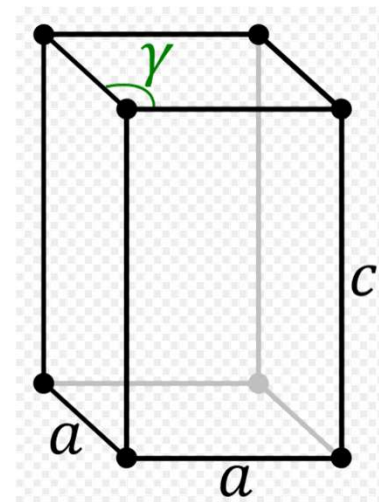
i -index is derived from the reciprocal of fractional intercept of the plane on a_3 axis

Sometimes hexagonal indices are written with i as a dot ($hk.i$) or omitted (hkl)



$$a = b \neq c$$

$$\alpha = \beta = 90^\circ \quad \gamma = 120^\circ$$



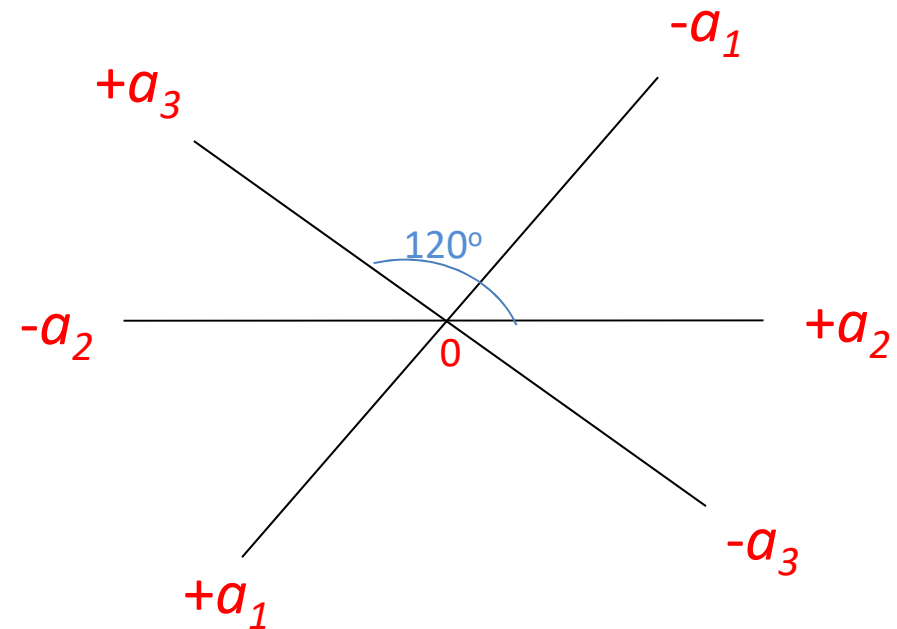
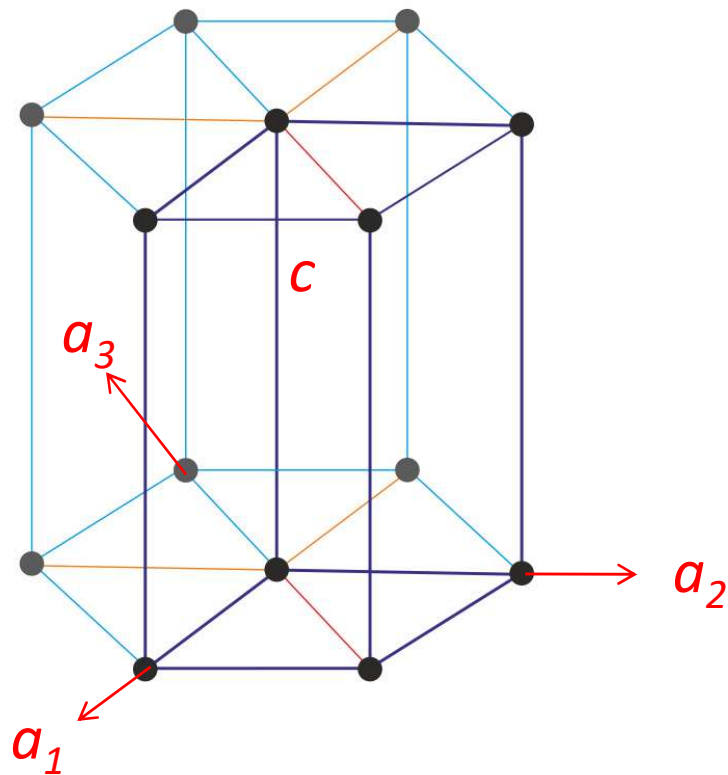
Hexagonal unit cell

Miller indices in hexagonal system

$(h\ k\ i\ l)$

$$h + k + i = 0$$

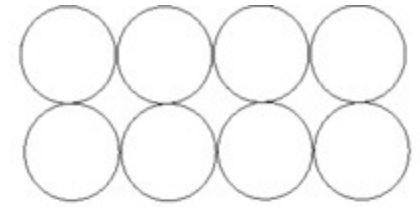
$$h + k = -i$$



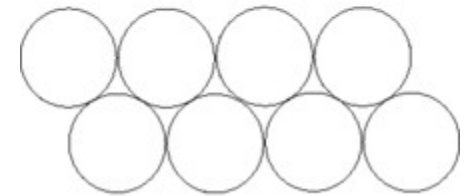
$(\bar{1}2.0)$

The other way to represent a solid

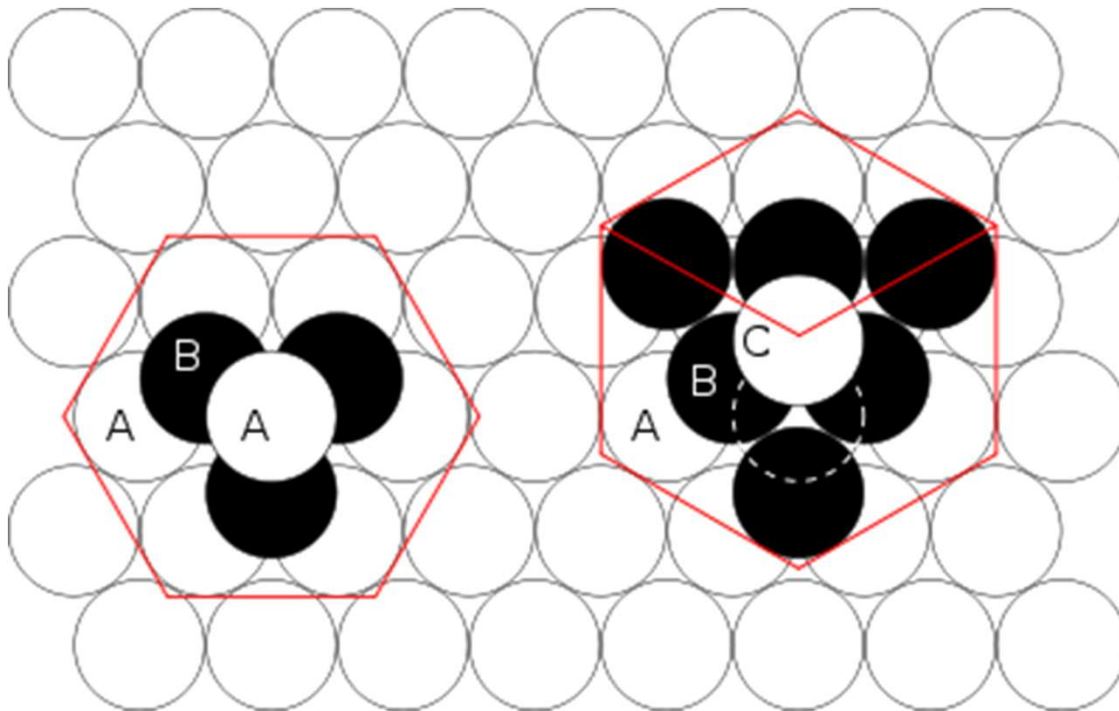
**Geometrical consideration : Assume atom to be a hard sphere
(don't overlap in space)**



Loose packing of sand grains



More compact packing



Hexagonal close packing (HCP)

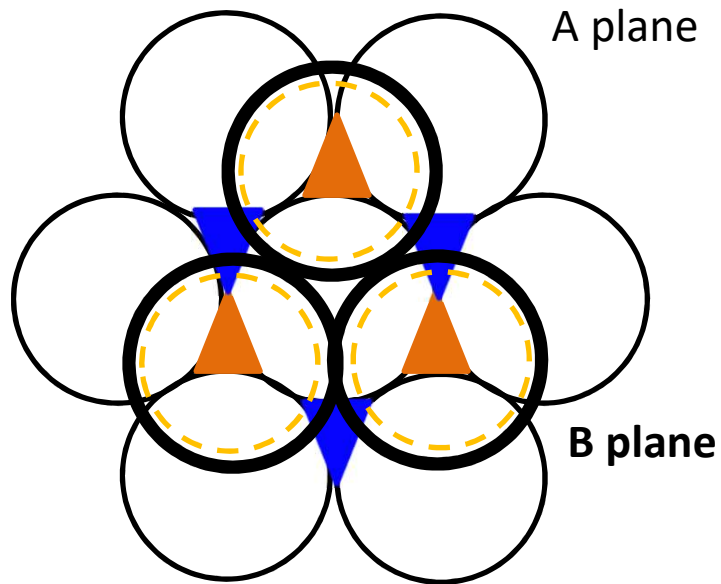
Cubic close packing (CCP)

The other way to represent a solid

Hexagonal close packing (HCP)

Cubic close packing (CCP)

Close packing **HOW TO DRAW**



...ABABAB... type packing

Above and top of B void

Bold (above the plane)

Dotted (below the plane)



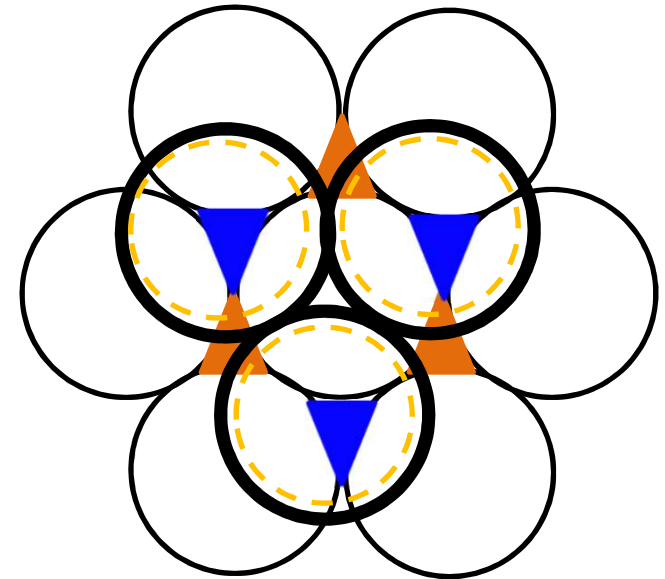
B void



C void

Above and top of C void

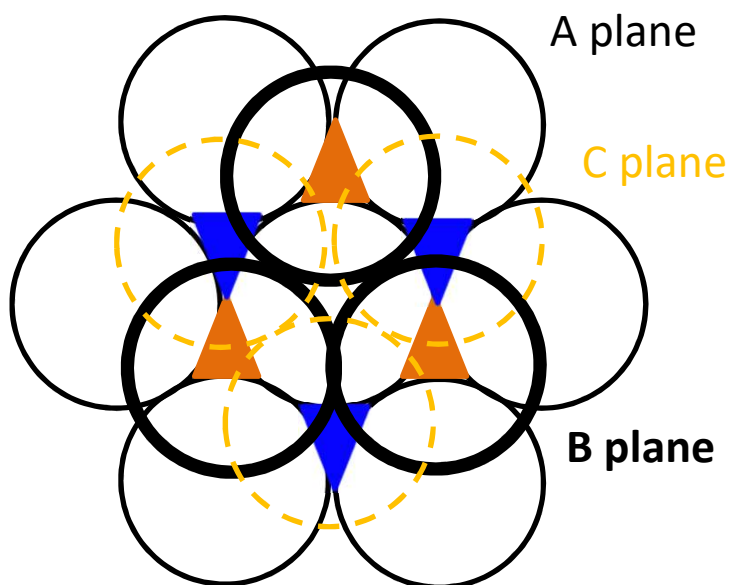
...ACACAC... type packing



The other way to represent a solid

Close packing **HOW TO DRAW**

Cubic close packing (CCP)



...ABCABC... type packing

Above on top of B void and below at C void

Bold (above the plane)

Dotted (below the plane)

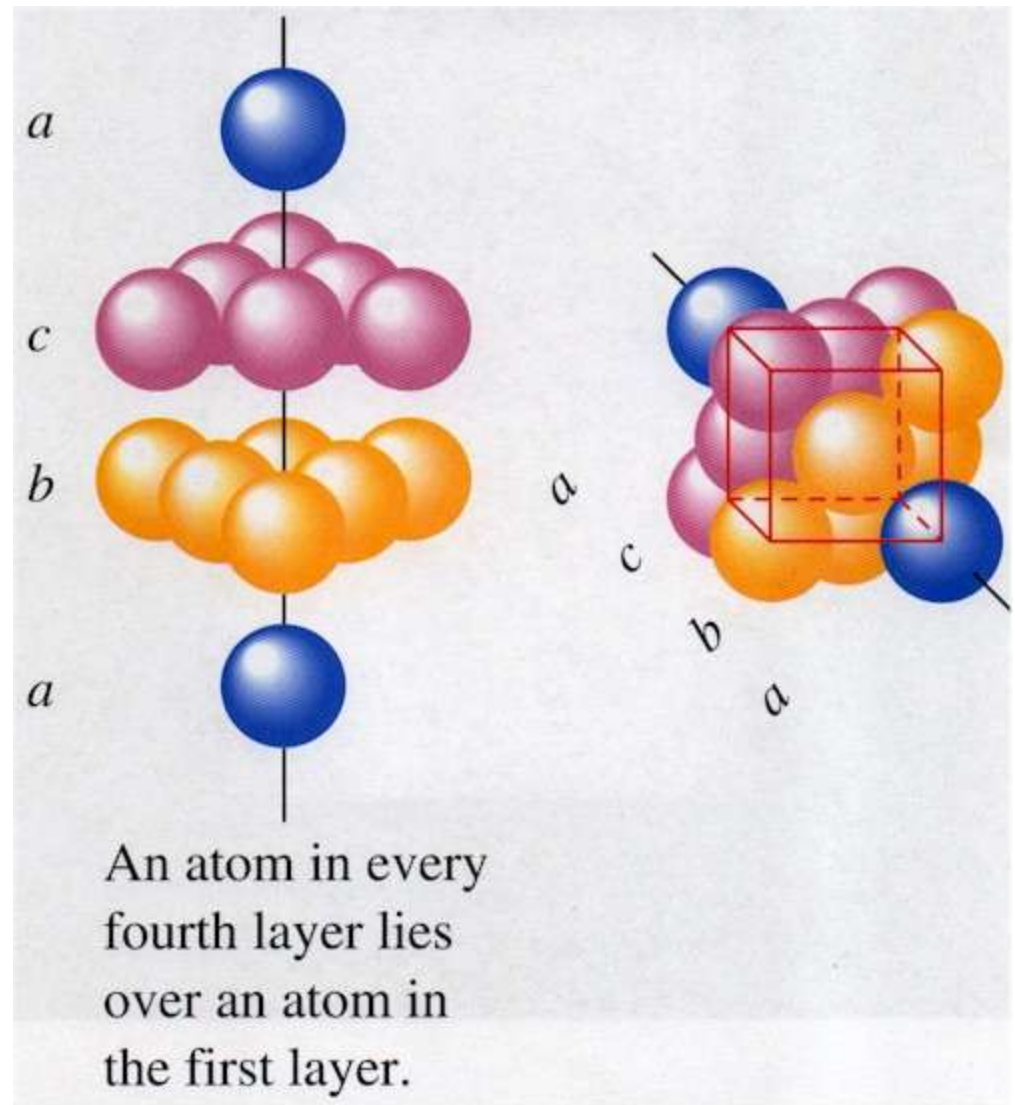
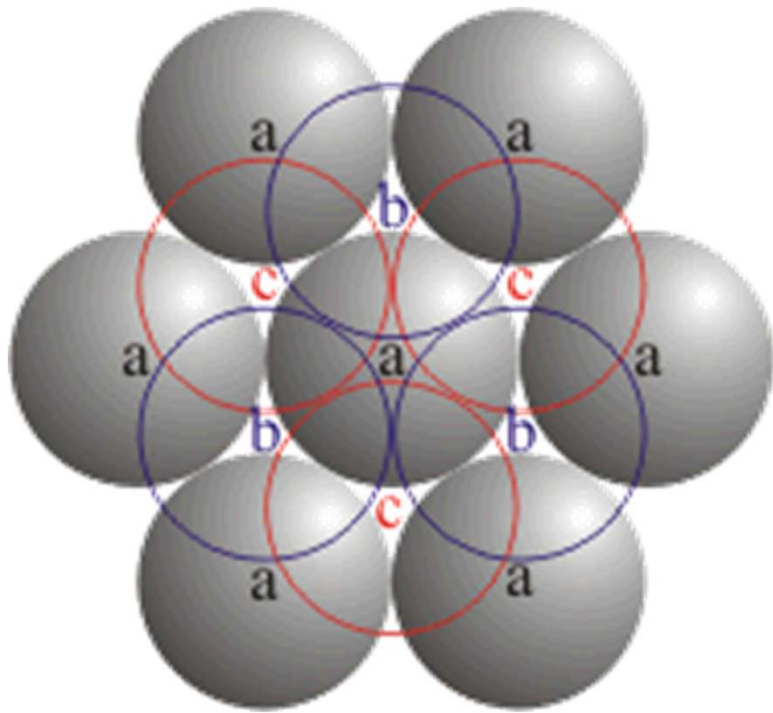


B void

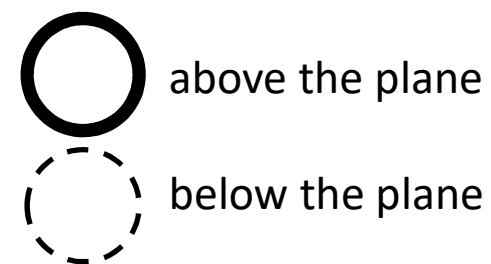


C void

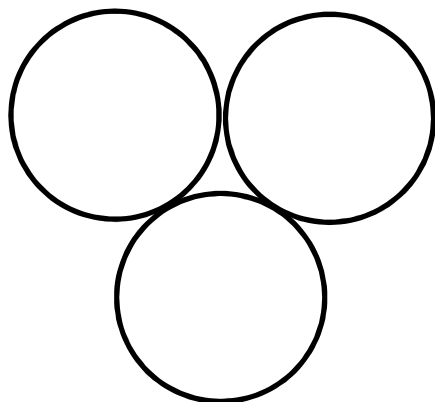
Close-packing



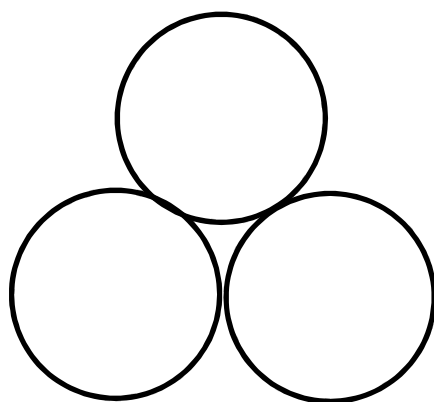
Voids



Trigonal hole

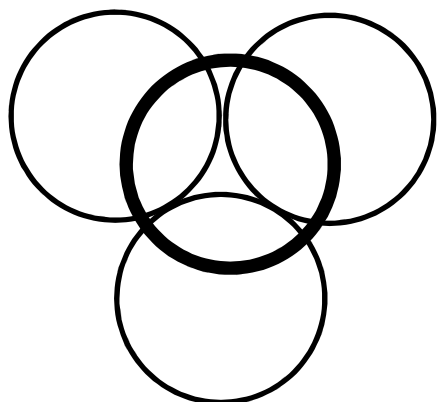


 T_+ site

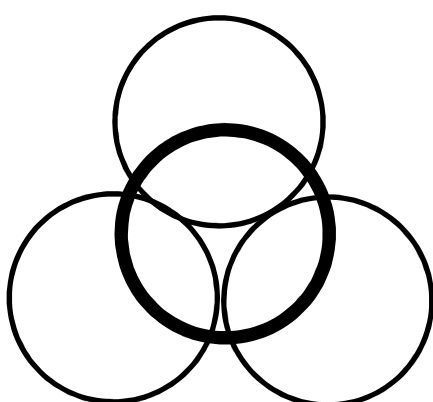


 T_- site

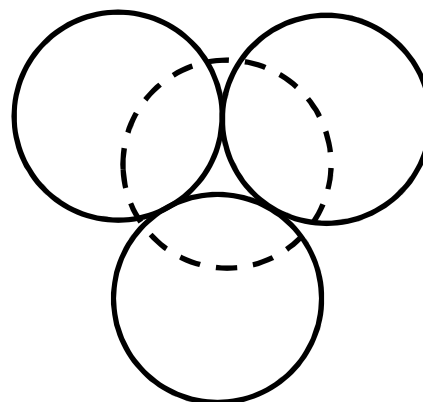
Tetrahedral hole



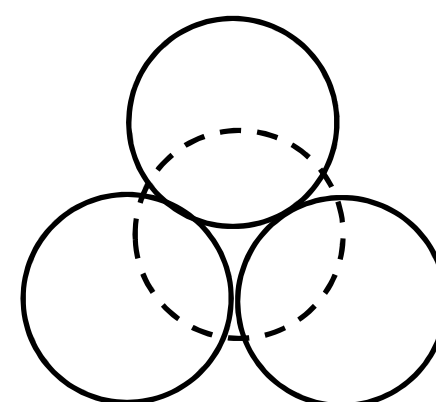
 T_+ site



 T_- site

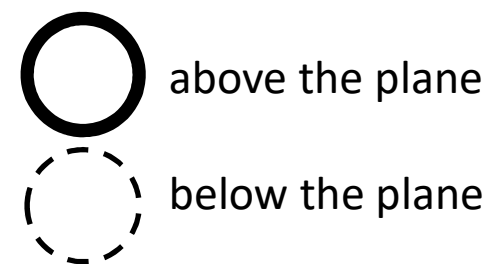


 T_- site

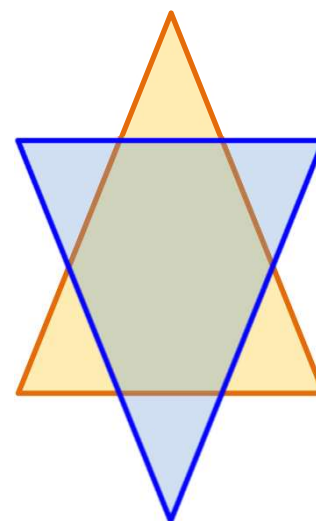
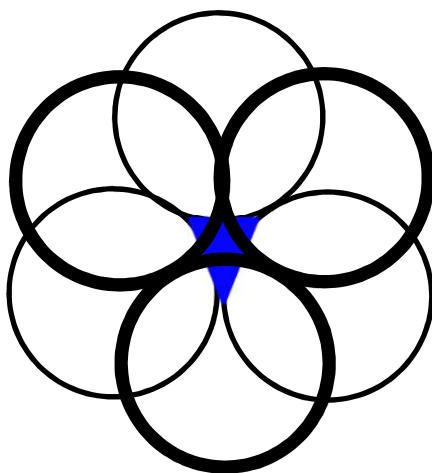


 T_+ site

Voids



Octahedral hole



Radius ratio rule

The empty void can take up another atom:

- (1) 2 different types of atoms in 2 layers
- (2) 2 different types of atoms in the same layer

In an ideally ionic crystal structure, the coordination number of ions are determined largely by the electrostatic considerations; cations surrounded by anions

- Relation between the radius, co-ordination number and the structural arrangement of the molecule.

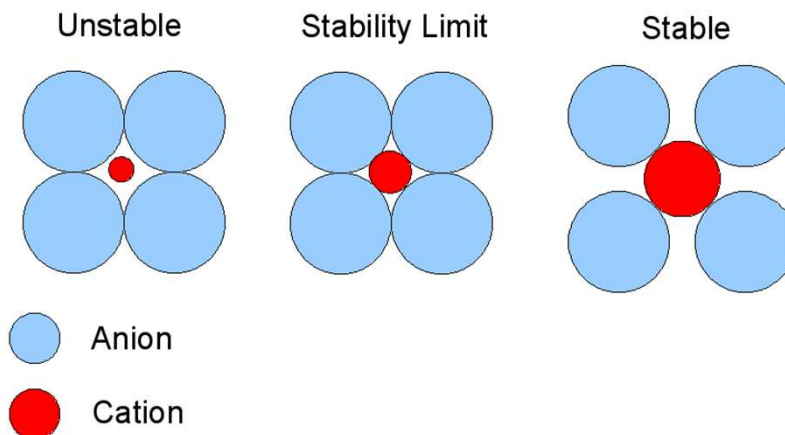
$$\text{Radius ratio} = \frac{\text{Radius of cation } (r^+)}{\text{Radius of anion } (r^-)}$$

- Greater the radius ratio, larger the size of the cation and hence the co-ordination number.

Coordination	Minimum r/R
Linear, 2	---
Trigonal, 3	0.155
Tetrahedral, 4	0.225
Octahedral, 6	0.414
Cubic, 8	0.732

For $r/R < 0.414$, the cation will not be in contact with all the nearest neighbor anions

For $r/R > 0.414$, the cation will push the anions apart and this happens up to $r/R = 0.732$



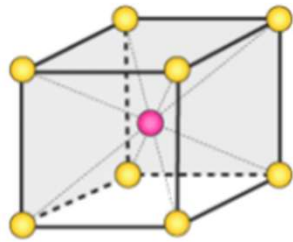
Radius ratio rule: homework

Derive radius ratio for:

Tetrahedral coordination

Octahedral coordination

Packing efficiency



Body-centred
Cubic Unit Cell
(BCC)

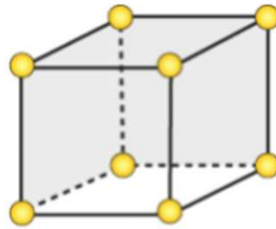


$$\text{Packing efficiency} = \frac{\text{Volume occupied by two spheres in the unit cell} \times 100}{\text{Total volume of the unit cell}} \%$$

$$= \frac{2 \times \left(\frac{4}{3}\right) \pi r^3 \times 100}{\left[\left(\frac{4}{\sqrt{3}}\right) r\right]^3} \%$$

$$= \frac{(8/3) \pi r^3 \times 100}{64 / (3\sqrt{3}) r^3} \%$$

$$= 68\%$$



Simple cubic



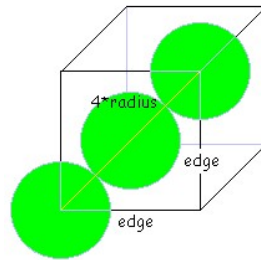
$$\text{Packing efficiency} = \frac{\text{Volume of one atom}}{\text{Volume of cubic unit cell}} \times 100\%$$

$$= \frac{\frac{4}{3} \pi r^3}{8r^3} \times 100$$

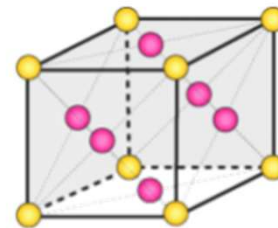
$$= \frac{\pi}{6} \times 100$$

$$= 52.36\%$$

$$= 52.4 \%$$



Packing Efficiency



Face-centred
Cubic Unit Cell
(FCC)

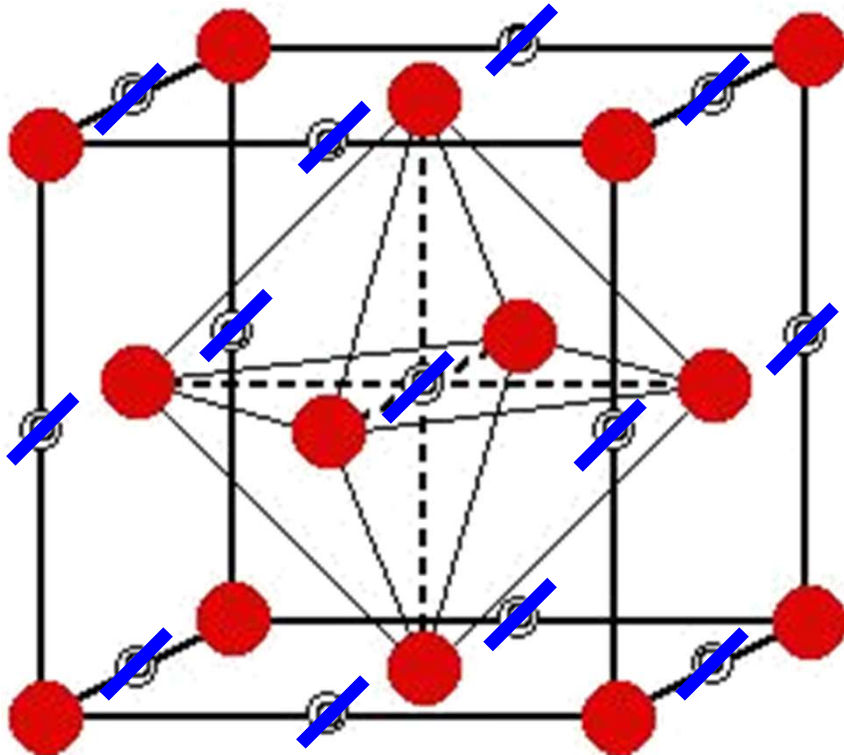
$$\text{edge} = 2\sqrt{2} * \text{radius}$$



$$\frac{4 \text{ atoms} * \text{volume per atom}}{\text{volume of unit cell}} \times 100$$

$$\frac{4 * \frac{4}{3} \pi r^3}{(2\sqrt{2} * r)^3} \times 100 = 74\%$$

Octahedral hole (oh) ✓



Each edge is shared by 4 unit cells
Total 12 octahedral holes

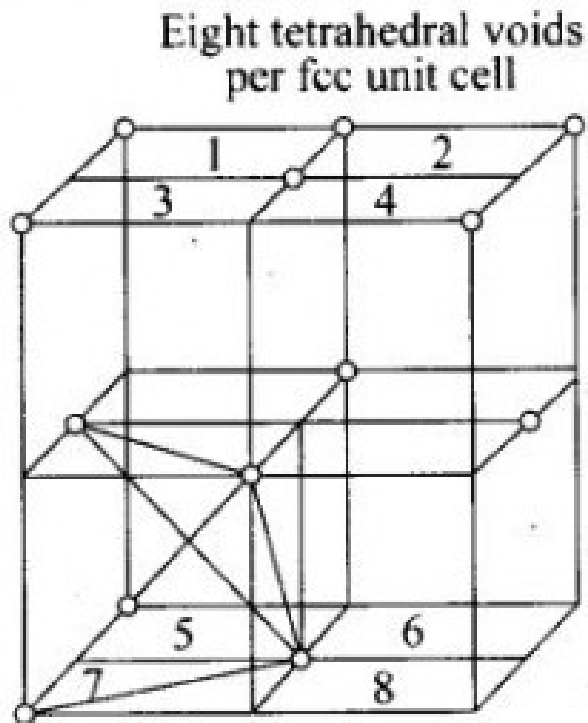
4 oh voids per unit cell

3 – edges ; 1 - center

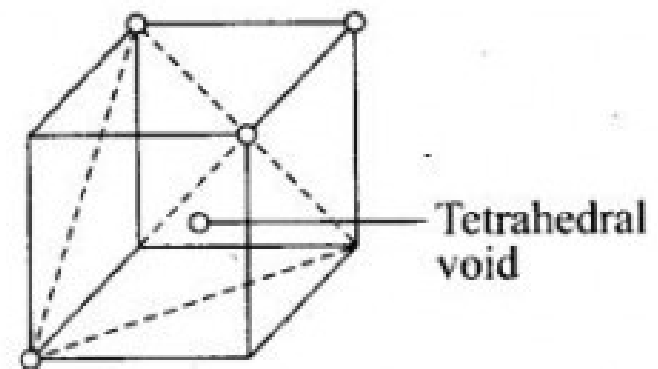
Coordinates:

$0 \frac{1}{2} 0$ $\frac{1}{2} 0 0$ $0 0 \frac{1}{2}$ $\frac{1}{2} \frac{1}{2} \frac{1}{2}$

Tetrahedral hole (td)

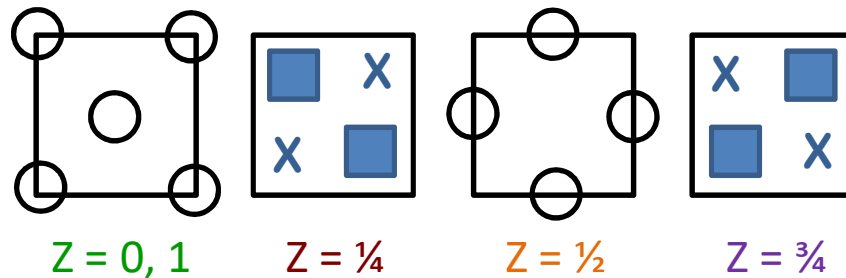
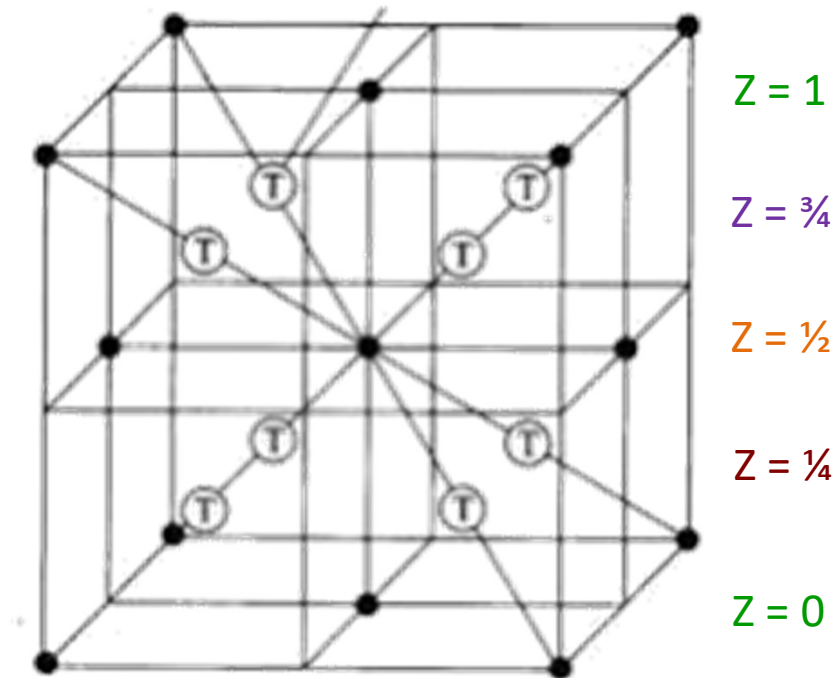


Each cube represented by numeric 1, 2, 3, 4, 5, 6, 7, 8 contains one tetrahedral void.



Each cube contains one tetrahedral void at its body centre as shown above

Tetrahedral hole (td)



In a FCC structure, shift everything along the body diagonal ($\frac{1}{4}$ th of the body diagonal)

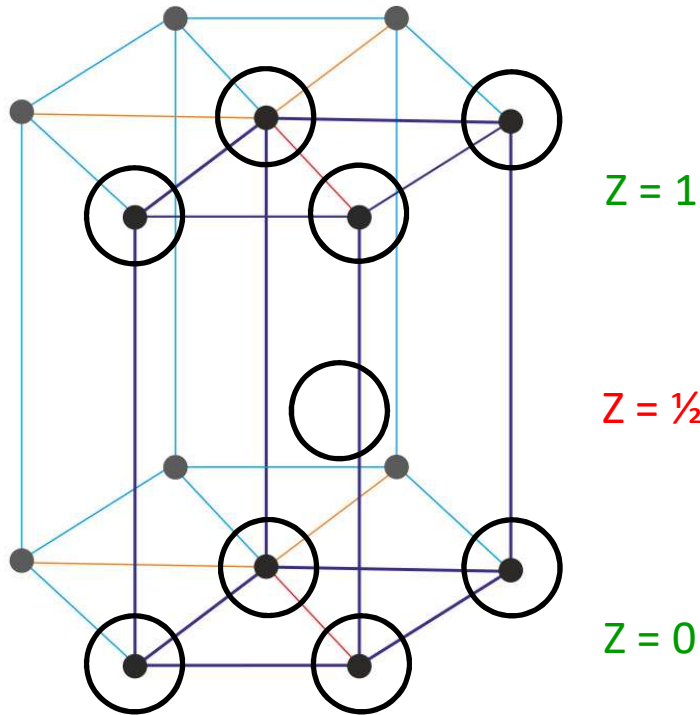
8 td voids per unit cell

At the center of 8 smaller cubes

Coordinates:

$0\ 0\ 0$	$\frac{1}{2}\ \frac{1}{2}\ 0$	$0\ \frac{1}{2}\ \frac{1}{2}$	$\frac{1}{2}\ 0\ \frac{1}{2}$	FCC
$\frac{1}{4}\ \frac{1}{4}\ \frac{1}{4}$	$\frac{3}{4}\ \frac{3}{4}\ \frac{1}{4}$	$\frac{1}{4}\ \frac{3}{4}\ \frac{3}{4}$	$\frac{3}{4}\ \frac{1}{4}\ \frac{3}{4}$	shift
$\frac{3}{4}\ \frac{3}{4}\ \frac{3}{4}$	$\frac{1}{4}\ \frac{1}{4}\ \frac{3}{4}$	$\frac{3}{4}\ \frac{1}{4}\ \frac{1}{4}$	$\frac{1}{4}\ \frac{3}{4}\ \frac{1}{4}$	

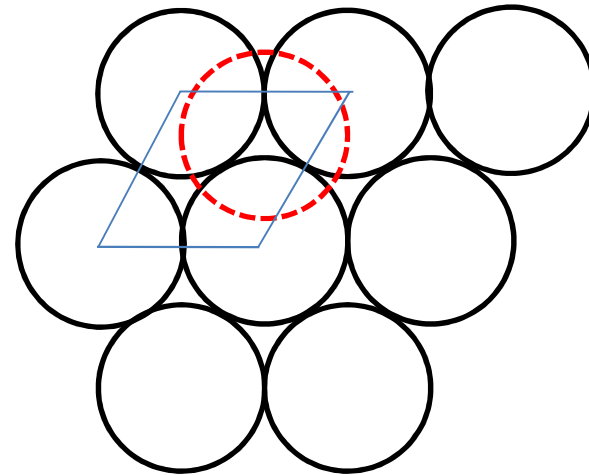
Hexagonal close packed (HCP) structure



The location of neighbours is the same, if the void above is Δ , the void below is also Δ

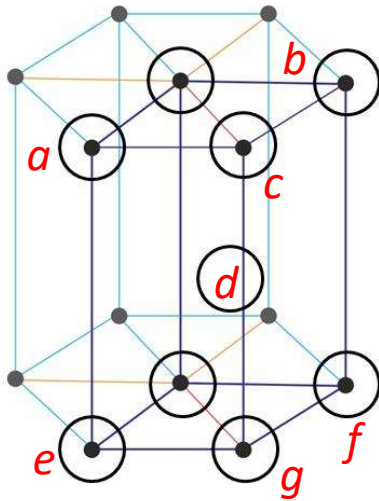
Basis (motif): 000 $\frac{1}{3}$ $\frac{2}{3}$ $\frac{1}{2}$

OR 000 $\frac{2}{3}$ $\frac{1}{3}$ $\frac{1}{2}$

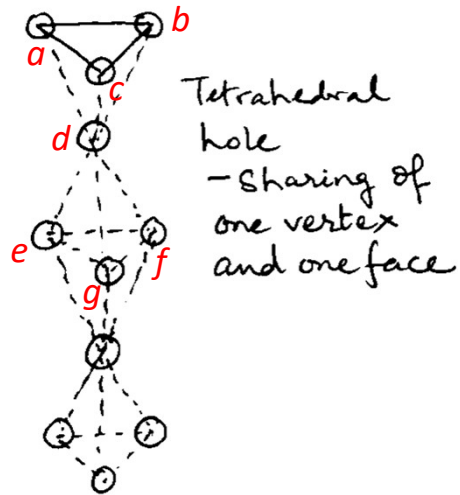


Ideal HCP lattice: $\frac{c}{a} = \sqrt{\frac{8}{3}} = 1.633$

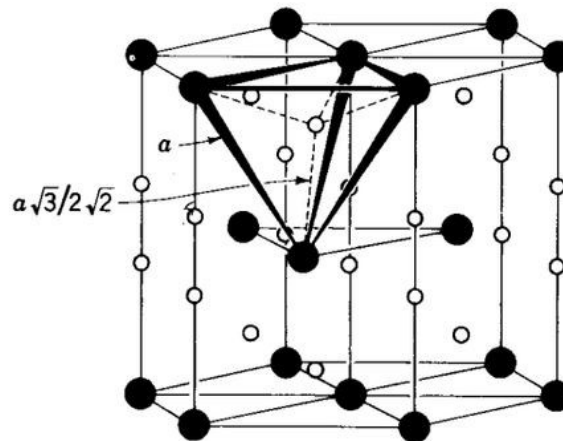
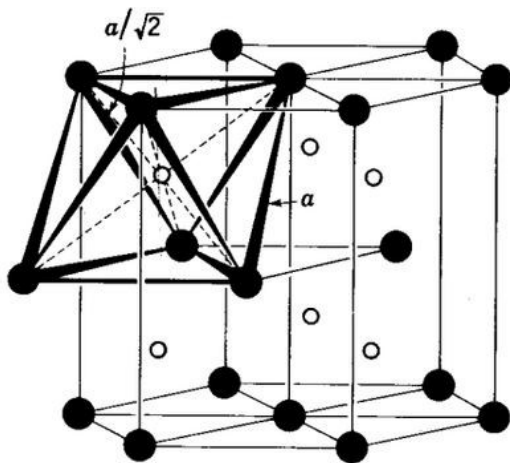
Voids in HCP structure



Oh voids



Td voids



Coordinates:

$$\text{Anions : } 000 \quad \frac{2}{3} \frac{1}{3} \frac{1}{2}$$

$$T_+ : \frac{2}{3} \frac{1}{3} \frac{1}{8} \quad 00 \frac{5}{8}$$

$$T_- : \frac{2}{3} \frac{1}{3} \frac{7}{8} \quad 00 \frac{3}{8}$$

$$O_h : \frac{1}{3} \frac{2}{3} \frac{1}{4} \quad \frac{1}{3} \frac{2}{3} \frac{3}{4}$$

The other way

$$\text{Anions : } 000 \quad \frac{1}{3} \frac{2}{3} \frac{1}{2}$$

$$T_+ : \frac{1}{3} \frac{2}{3} \frac{1}{8} \quad 00 \frac{5}{8}$$

$$T_- : \frac{1}{3} \frac{2}{3} \frac{7}{8} \quad 00 \frac{3}{8}$$

$$O_h : \frac{2}{3} \frac{1}{3} \frac{1}{4} \quad \frac{2}{3} \frac{1}{3} \frac{3}{4}$$